Final Project: Windows and Linux Tasks

Lab Report

Cyber Secure Inc. — Junior Cybersecurity Analyst
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Overview

Cyber Secure Inc. handles system administration and security for contracted businesses. As a junior cybersecurity analyst, I was assigned six tickets covering both Windows and Linux administration. This report documents the completion of the three Windows tasks assigned in Part 1: creating a new user and group, updating and running virus and threat protection, and configuring the Windows Defender Firewall, along with the three Linux tasks assigned in Part 2: creating a new user, managing files and folders, and applying system updates.

Environment	Microsoft Windows Server and Linux terminal (Coursera lab environments)
Tools Used	Server Manager, Computer Management, Windows Security, Windows Defender Firewall with Advanced Security, Linux command line (adduser, mkdir, touch, apt, less)
Tasks Completed	6 of 6 tickets (3 Windows, 3 Linux)

Ticket 1: Create a New User and Group

Objective: *Create a new user in Windows Server Manager, create a new group named "Accounting," and add the new user to that group.*

Step 1 — Create the New User Account

Using Computer Management (Local) under Server Manager, I navigated to System Tools > Local Users and Groups > Users and selected "New User." I created an account named after a favorite character, "Spiderman" (full name: Peter Parker), and set a secure password with the "User must change password at next logon" option enabled.

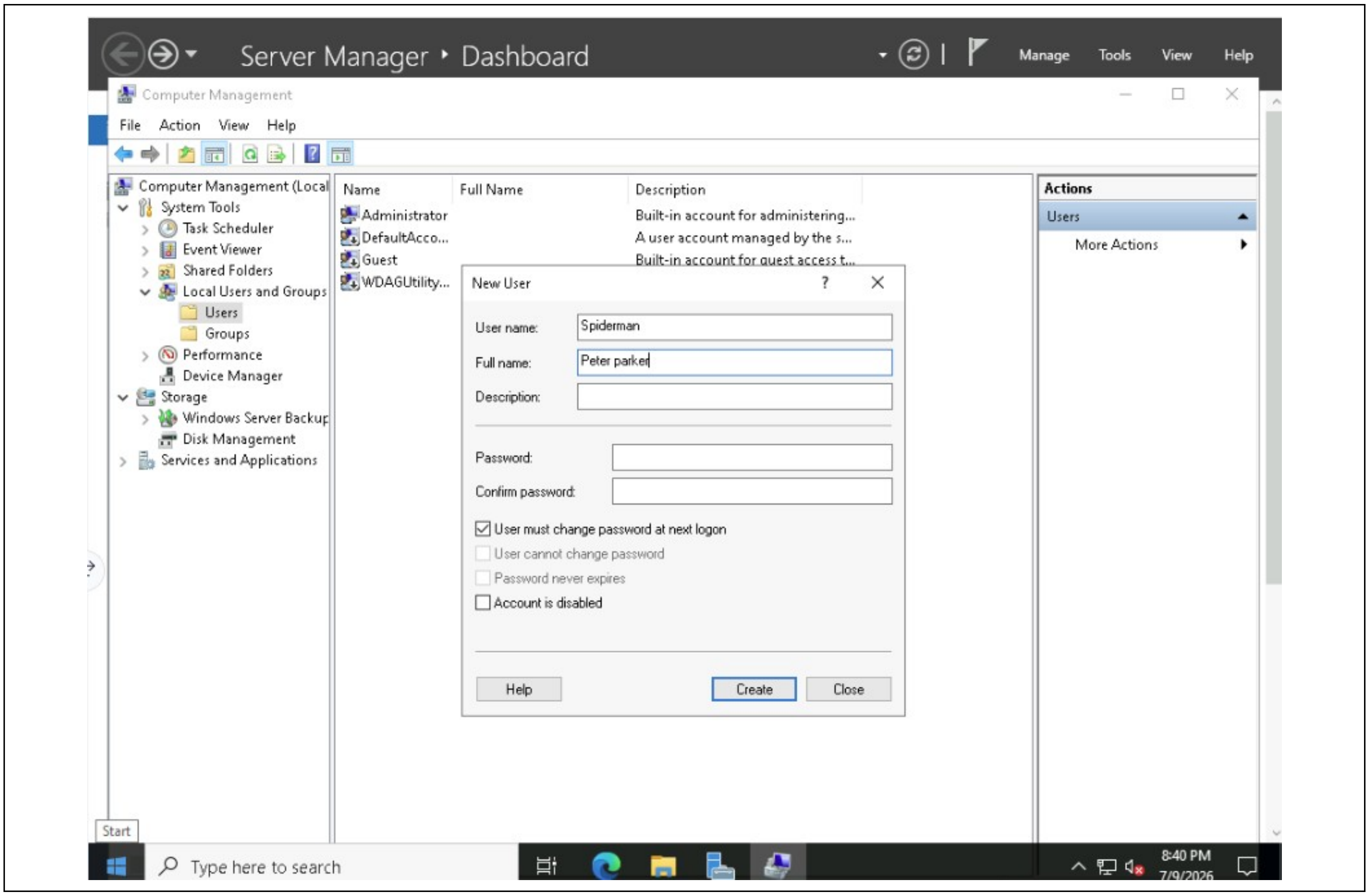

Figure 1. New User dialog in Server Manager showing username "Spiderman" and full name "Peter Parker."

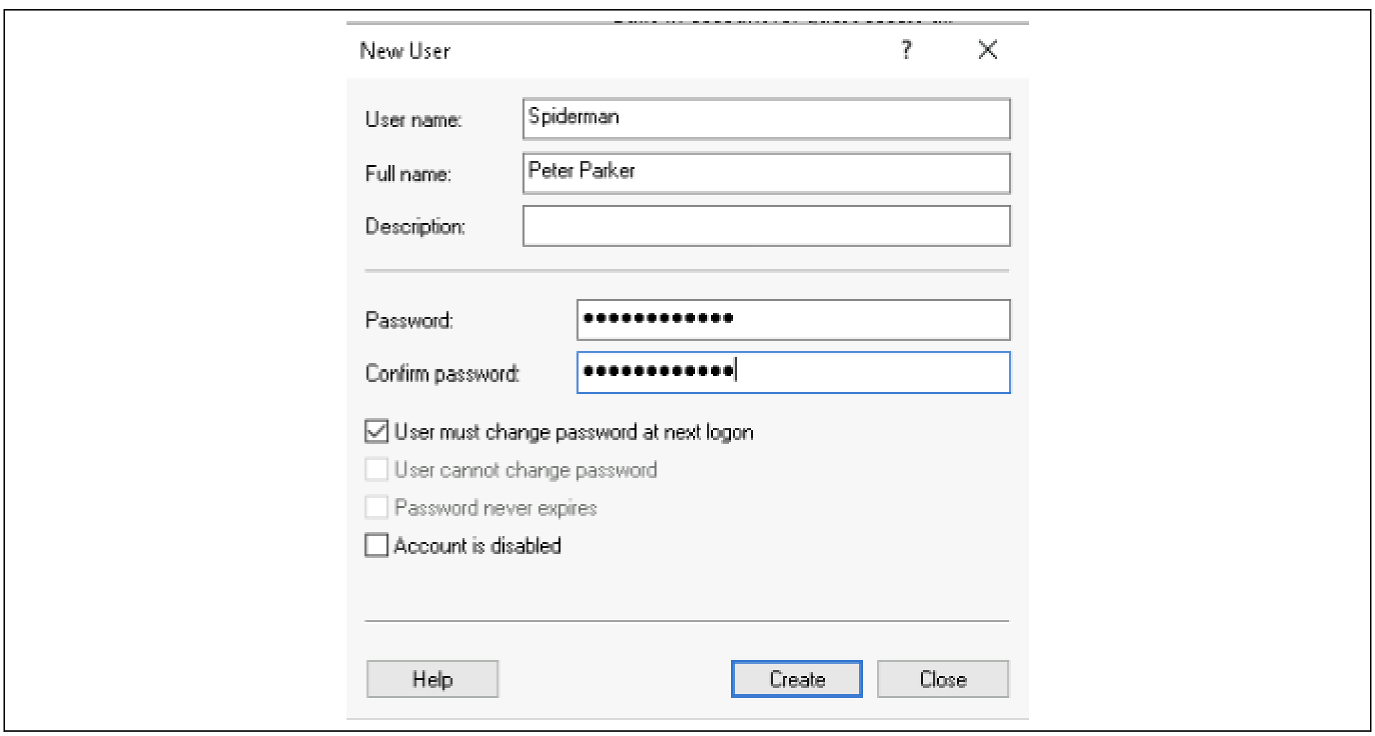

Figure 2. New User dialog with a secure password entered and confirmed prior to account creation.

Step 2 — Create the Accounting Group

Next, I navigated to Local Users and Groups > Groups and created a new group named "Accounting." The screenshot below shows the Accounting group listed among the built-in Windows groups in Computer Management.

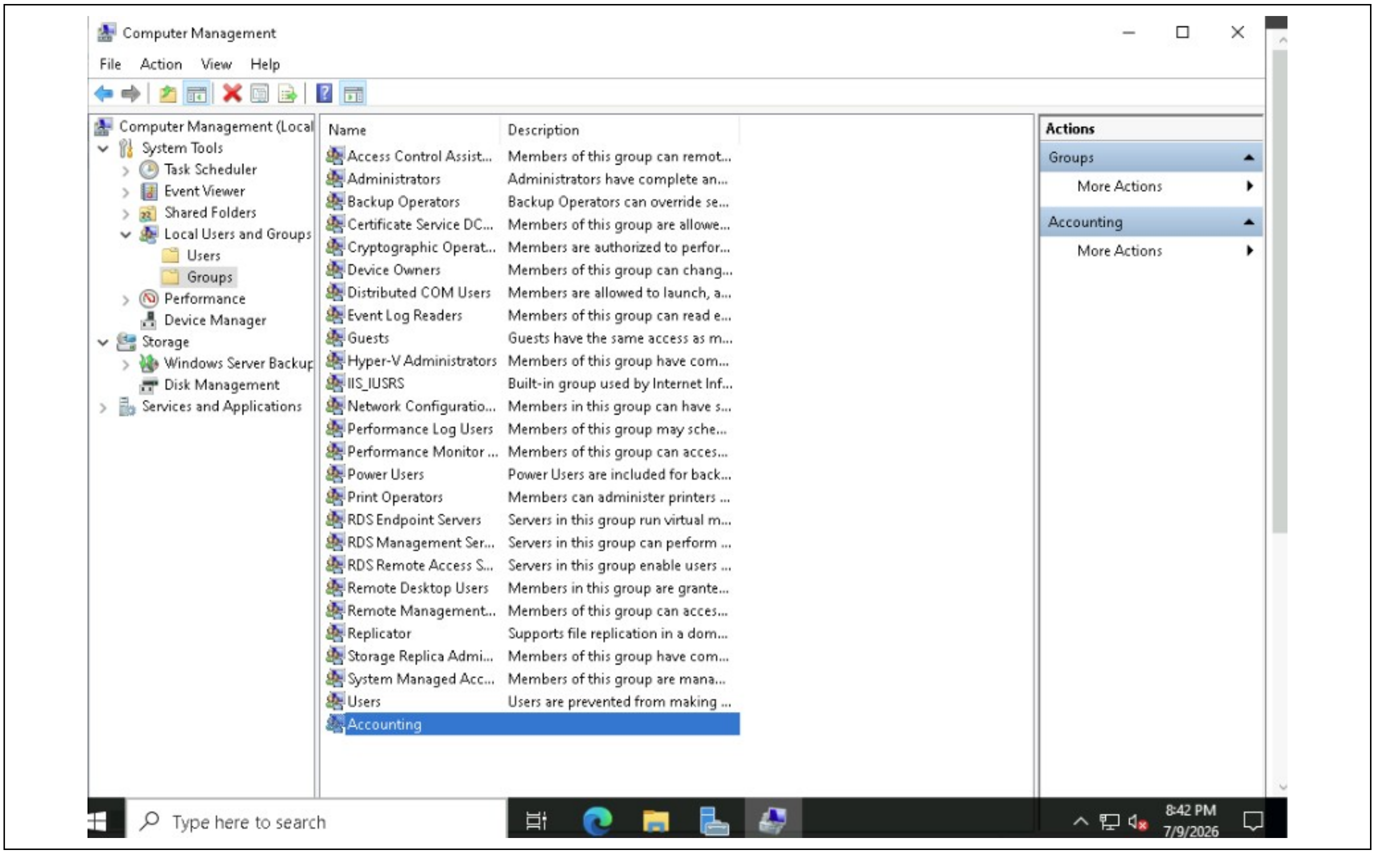

Figure 3. Computer Management showing the newly created "Accounting" group in the Groups list.

Step 3 — Add the User to the Group

I opened the Accounting group's Properties and added the "Spiderman" user account as a member. The screenshot below confirms that Spiderman now appears in the Accounting group's Members list.

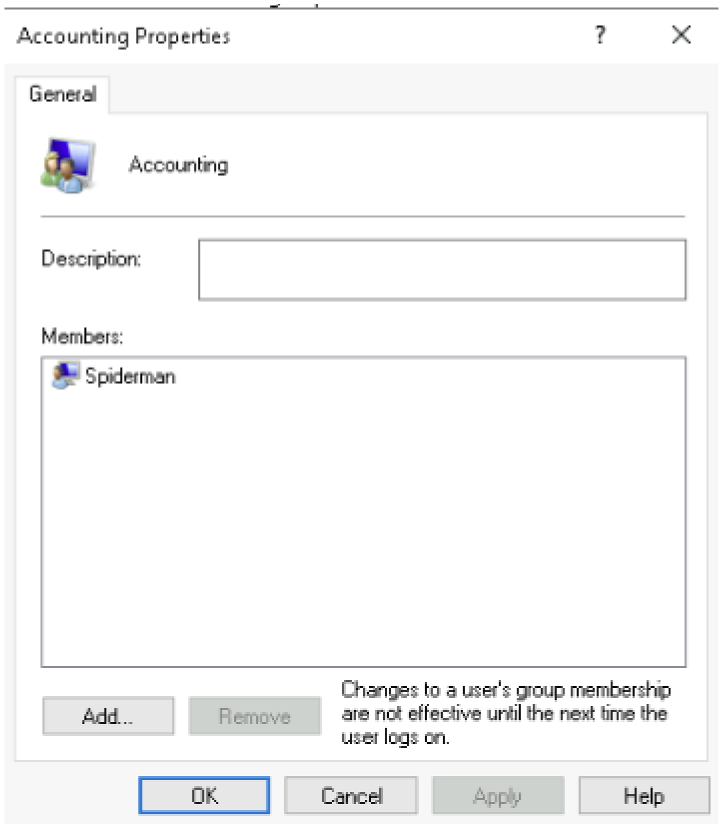

Figure 4. Accounting Properties dialog showing "Spiderman" listed as a member of the group.

Ticket 2: Check for Virus and Threat Protection Updates

Objective: *Check for updates to Windows Defender virus and threat protection and run a Quick Scan to confirm the system is protected.*

Step 1 — Check for Protection Updates

In Windows Security, I opened Virus & Threat Protection > Protection Updates and selected "Check for updates." The system confirmed the security intelligence was up to date, with the update applied on 7/9/2026 at 8:40 PM.

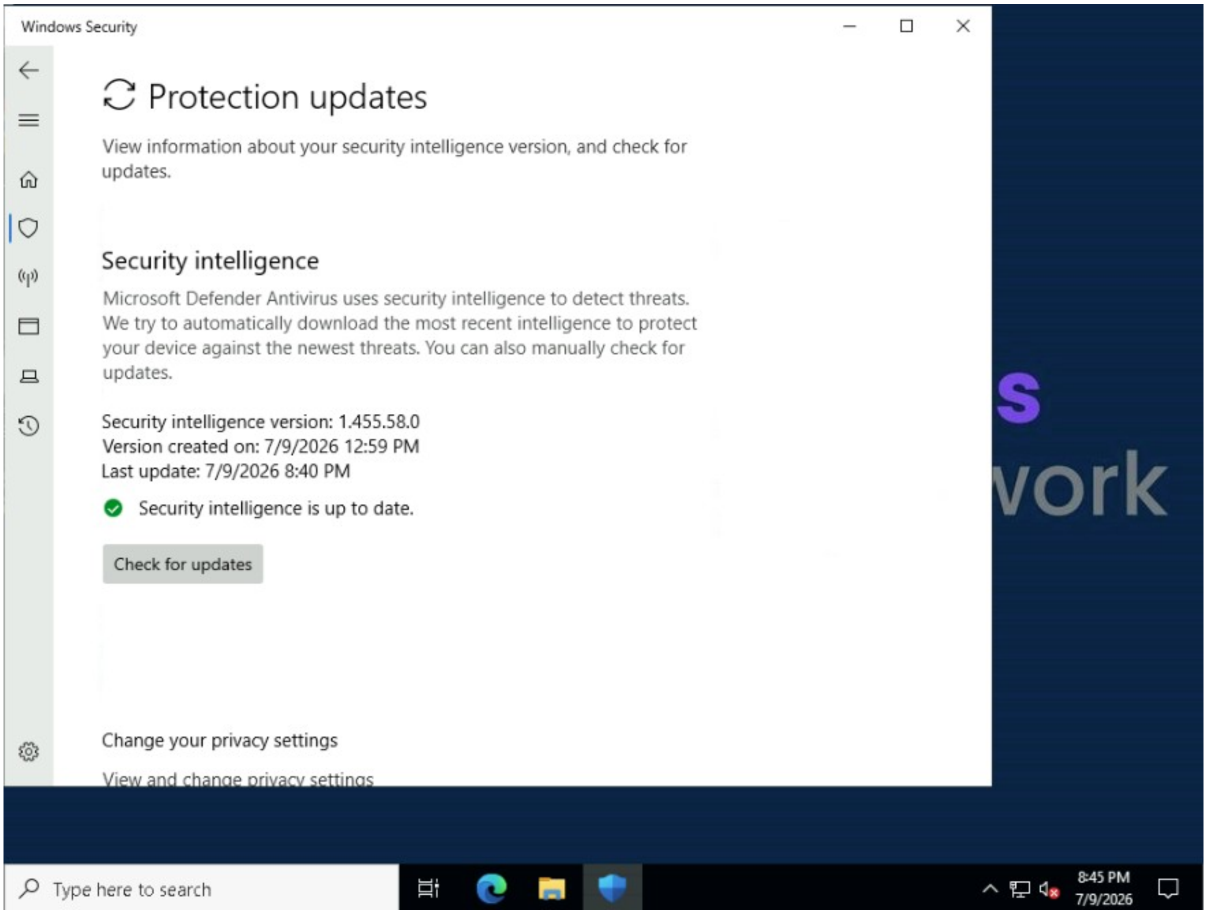

Figure 5. Protection Updates screen confirming security intelligence version 1.455.58.0, last updated 7/9/2026 8:40 PM.

Step 2 — Run a Quick Scan

I returned to the main Virus & Threat Protection screen and ran a Quick Scan. The scan completed successfully with 0 threats found across 29,294 files scanned, completed on 7/9/2026 at 8:45 PM.

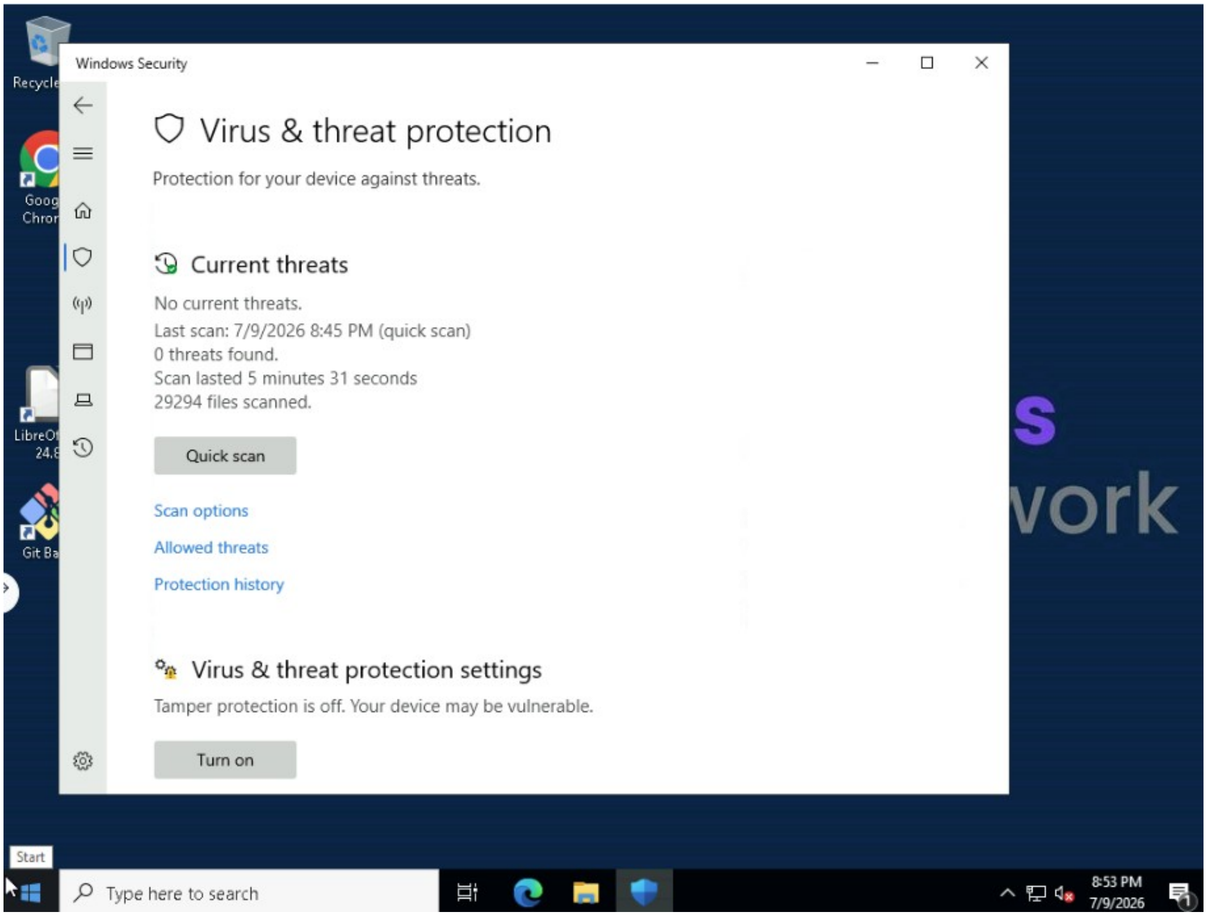

Figure 6. Virus & Threat Protection screen showing the Quick Scan completed on 7/9/2026 8:45 PM with 0 threats found.

Ticket 3: Configure Firewall and Network Protection

Objective: *Create a new inbound rule in Windows Defender Firewall with Advanced Security that controls TCP connections on port 80, applies to the Domain, Private, and Public profiles, and is named "Port 80 Permitted."*

I opened Windows Defender Firewall with Advanced Security and created a new inbound rule for TCP port 80, allowing the connection across all three network profiles (Domain, Private, and Public). The rule was named "Port 80 Permitted" as required. The Advanced tab of the rule's properties, shown below, confirms the profile scope and the action to allow the connection.

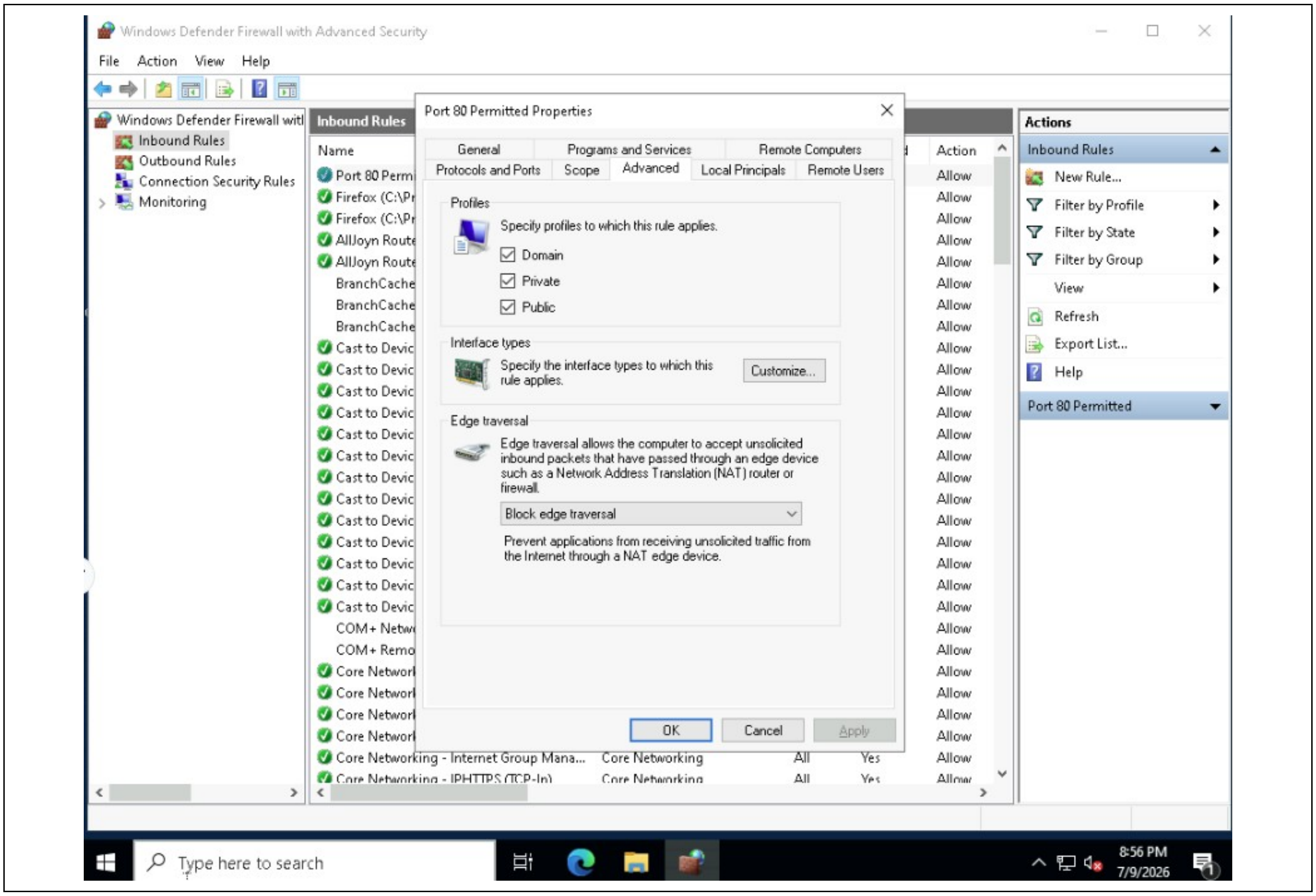

Figure 7. "Port 80 Permitted" inbound rule properties, showing the rule applied to the Domain, Private, and Public profiles and listed with an Allow action in the Inbound Rules list.

Ticket 4: Create a New User (Linux)

Objective: *Use the Linux terminal to create a new user with the adduser command, set a secure password and identifying information, and confirm the account was created by querying /etc/passwd.*

I ran adduser spiderman from the Linux terminal, named after the same favorite character used in the Windows portion of this project. I set a secure password and filled in the prompted full name and other identifying information, confirming it was correct. I then confirmed the account with cat /etc/passwd | grep spiderman, which returned the new account's UID, GID, full name, home directory, and default shell.

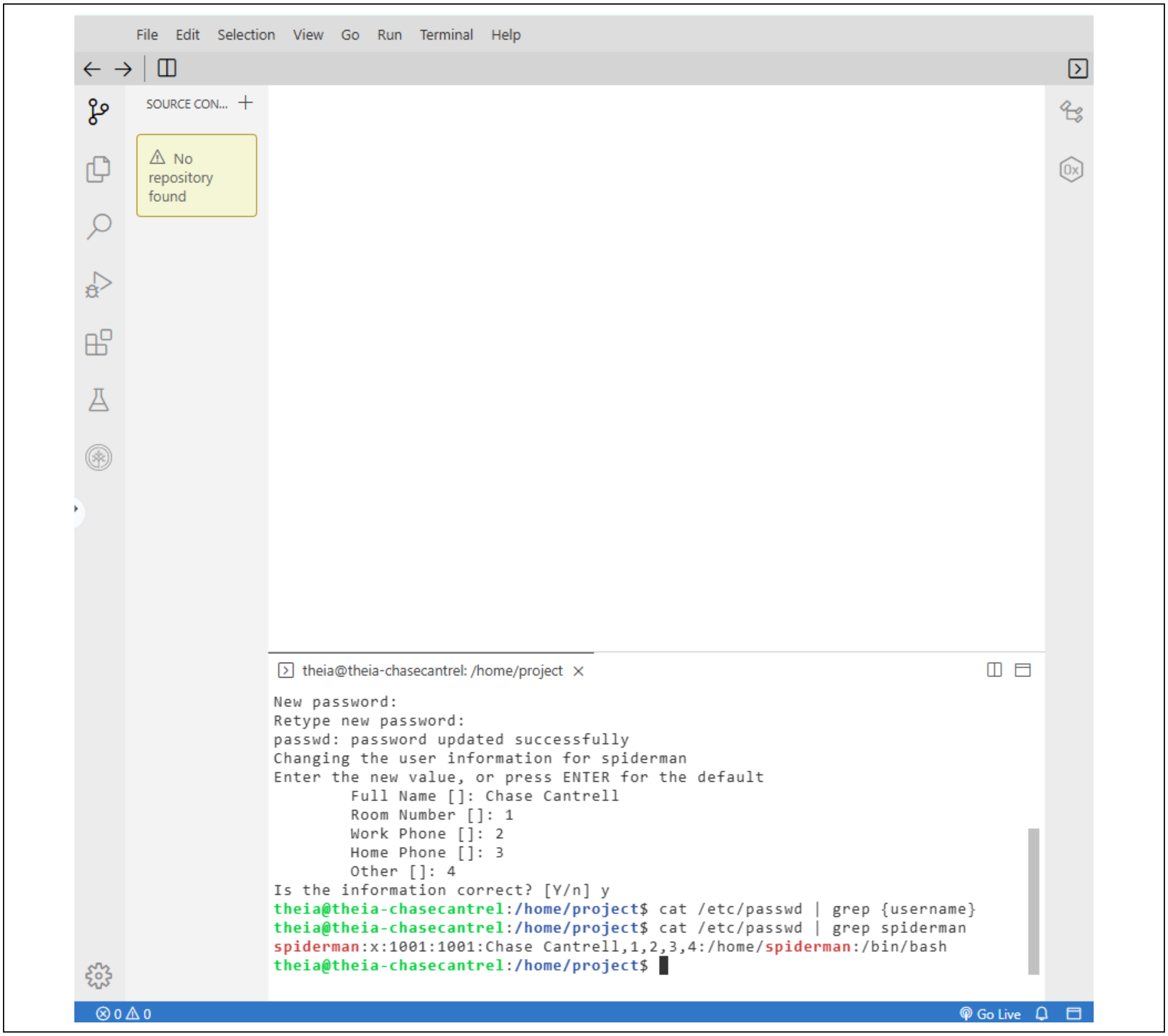

Figure 8. Terminal output showing the password set for user "spiderman," the full name "Chase Cantrell" entered at the adduser prompts, and the confirmed account entry from /etc/passwd.

Ticket 5: Manage Files and Folders (Linux)

Objective: *Create Accounting and Payroll folders, confirm their creation, create a Best Practices file inside the Accounting folder, and display the folder's contents.*

Step 1 — Create and Display the Folders

I created the two required folders with mkdir Accounting Payroll, then ran ls to confirm both folders were created in the home directory alongside an existing Untitled-1 file.

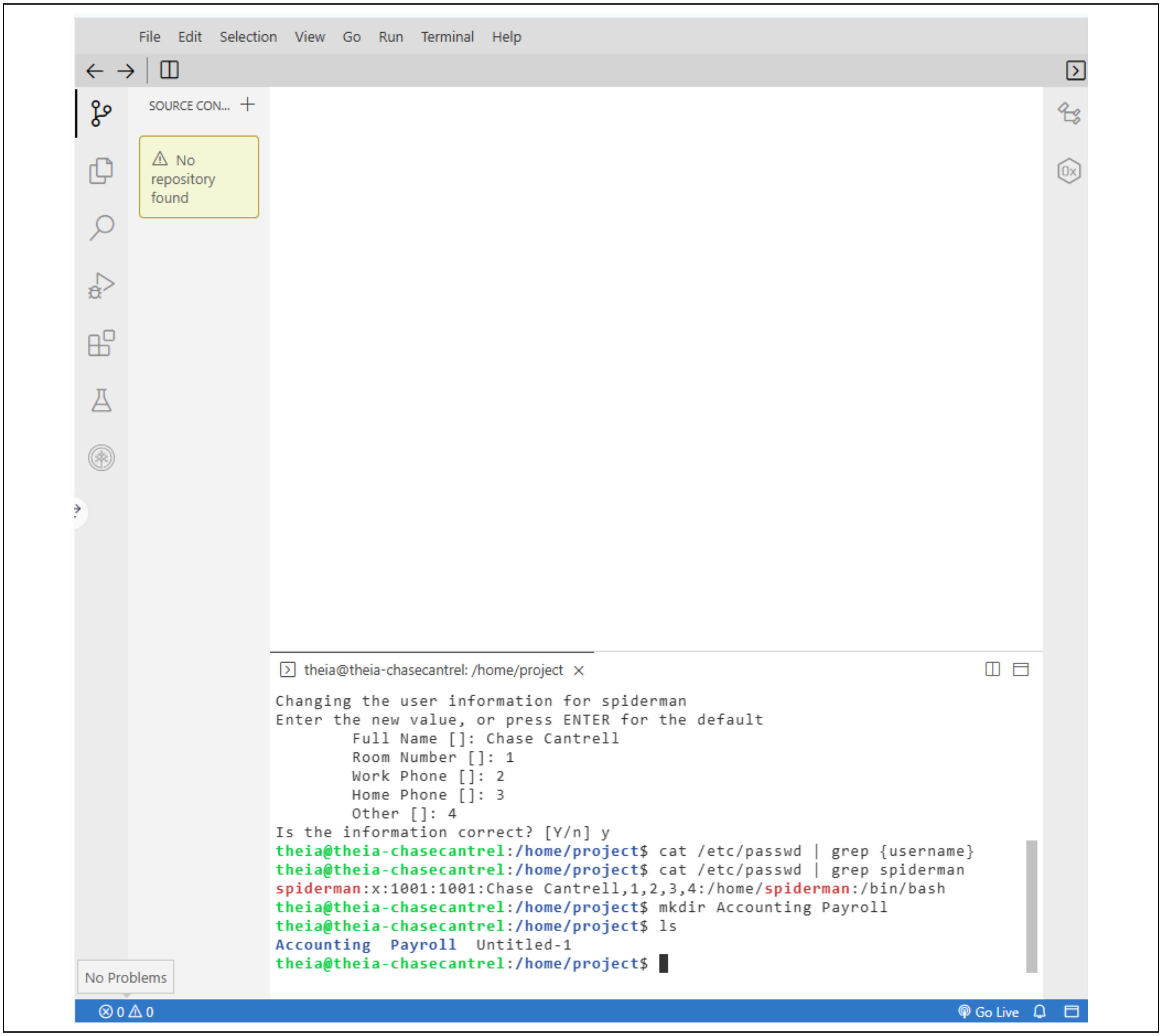

Figure 9. Terminal output showing the mkdir Accounting Payroll command and the resulting ls listing with both new folders.

Step 2 — Create the Best Practices File and Display Folder Contents

I created a file named "Best Practices" inside the Accounting folder using touch "Accounting/Best Practices," then ran ls Accounting to display the folder's contents and confirm the file was created successfully.

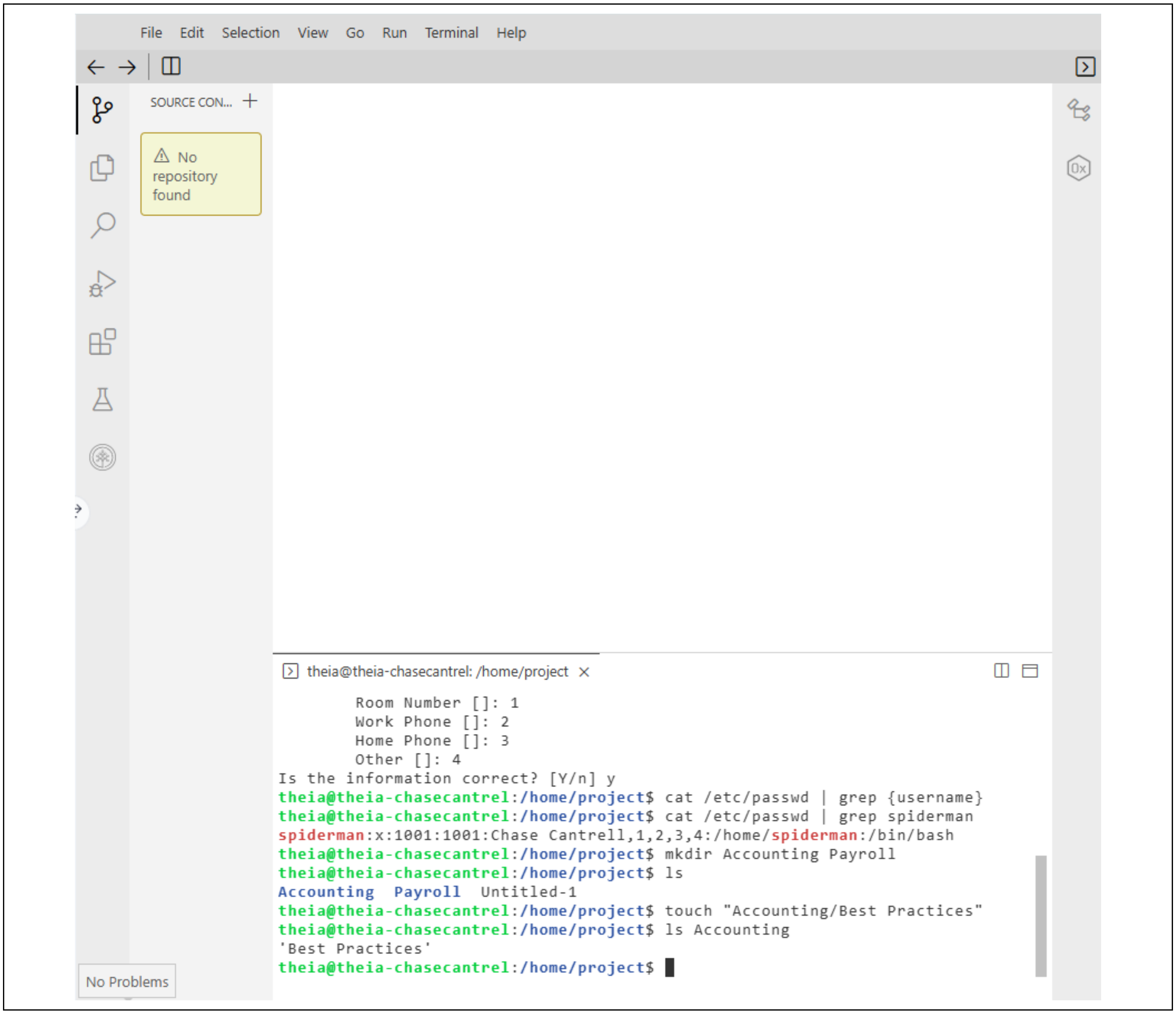

Figure 10. Terminal output showing the touch "Accounting/Best Practices" command and the ls Accounting listing confirming the file exists inside the folder.

Ticket 6: Apply System Updates (Linux)

Objective: *Check for and install system updates using apt, then review the update log with less /var/log/dpkg.log to confirm recent updates were applied.*

I checked for available packages with sudo apt update, then installed them with sudo apt upgrade. To verify the updates, I opened the package log with less /var/log/dpkg.log. The log below shows a series of package installation and configuration events completed on 2026-03-22, including the base-passwd and base-files packages moving through the unpacked, half-configured, and installed states.

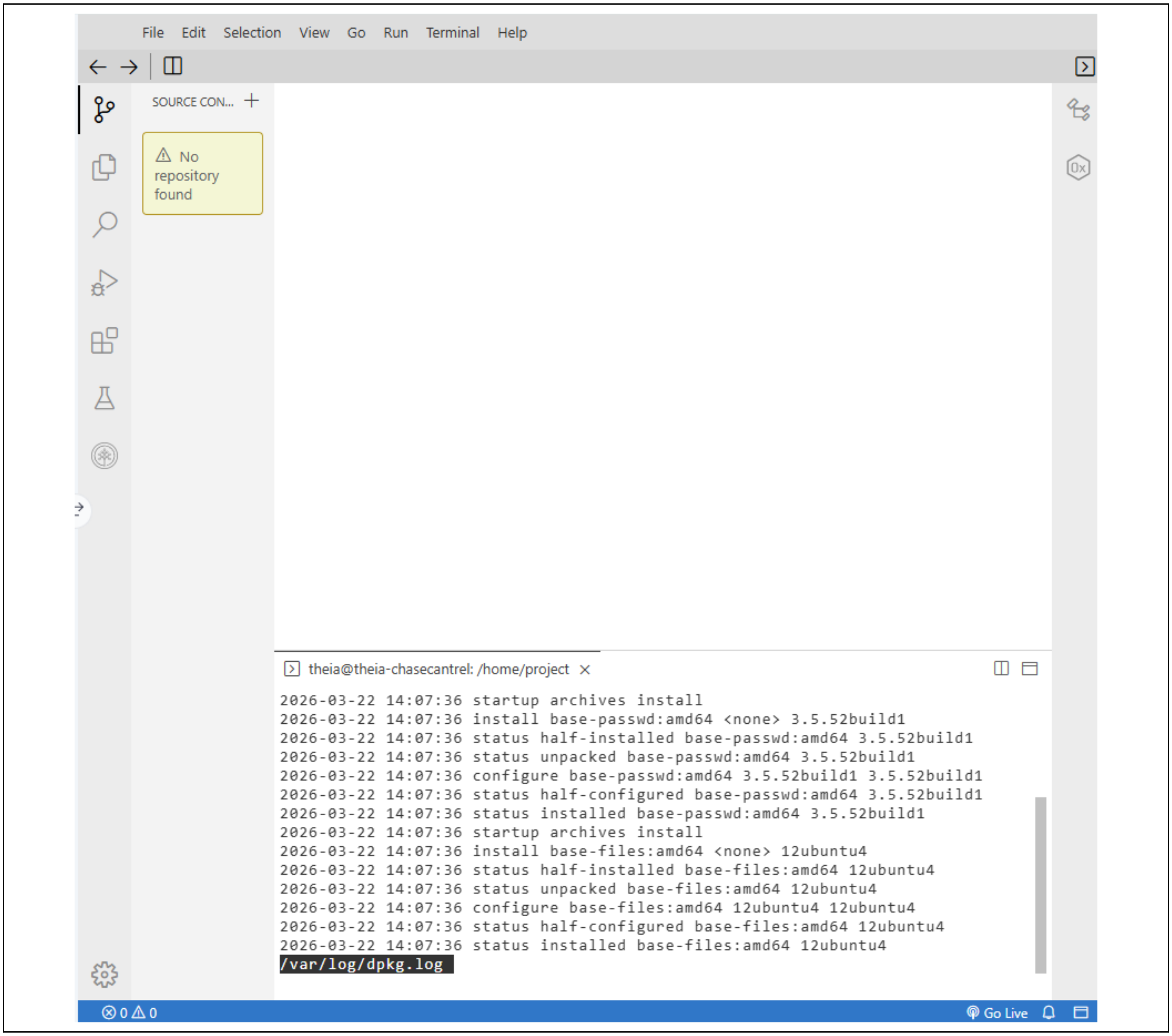

Figure 11. /var/log/dpkg.log contents displayed with less, showing recent package installation and configuration entries for base-passwd and base-files.

Conclusion

All six tickets across both the Windows and Linux portions of this project were completed successfully. On Windows, I created a new local user account and an Accounting group, added the user to that group, verified that Windows Defender's virus and threat protection definitions were current, ran a clean Quick Scan, and configured an inbound firewall rule permitting TCP traffic on port 80 across all network profiles. On Linux, I created a new user account with adduser and confirmed it through /etc/passwd, created and populated the Accounting and Payroll folders, and checked for and reviewed system updates using apt and the dpkg log. Together, these tasks demonstrate core system administration and security skills across both major operating system families, including user and group management, endpoint protection maintenance, firewall configuration, file system management, and patch management.